

Department of Leather Engineering
Khulna University of Engineering & Technology
 B. Sc. Engineering 3rd Year 1st Term Examination - 2022
LE 3109

Leather Products Engineering I

Time: 3 Hours.

Full Marks: 210

- N.B. i) Answer any THREE questions from each section in separate scripts.
 ii) Figures in the right margin indicate full marks.
 iii) Assume reasonable data if any missing.

SECTION-A

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|---|----|
| 1(a) Why allowances are important for leather goods manufacturing? | 06 |
| 1(b) What is clicking? State the qualities are required for a perfect clicker. | 08 |
| 1(c) Write the classification of leather goods according to volume and degree of reinforcement with example. | 11 |
| 1(d) Distinguish between underlay and overlay allowances. | 05 |
| 1(e) Mention the factors while pattern placement onto the leather before cutting. | 05 |
| 2(a) What is perspective drawing? Sketch the perspective view of a Bi-fold wallet. | 08 |
| 2(b) State the norms of pattern placement for leather products. | 08 |
| 2(c) Enumerate the leather consumption of a complete Bi-fold wallet. | 12 |
| 2(d) What is lining materials? Write down the objectives of lining materials for leather products. | 07 |
| 3(a) "Fashion constantly evolves with time"- Explain. | 07 |
| 3(b) Describe the properties of cutting board for leather. | 05 |
| 3(c) Describe the following operations during fabrication of a wallet: (i) Cradle construction of credit card pocket (ii) Centre piece. | 10 |
| 3(d) What are the differences among interlining, interfacing and under lining? | 09 |
| 3(e) Make the differentiation between pattern area and scale area for pattern placement. | 04 |
| 4(a) State the chronological development of fashion trends worldwide and their impression on the leather products industry. | 10 |
| 4(b) What are the differences among first, second and third wastages for leather products? | 07 |
| 4(c) "Factors should be considered during material consumption in a leather products industry"-Explain. | 08 |
| 4(d) How can you calculate leather wastages during cutting of material? | 04 |
| 4(e) State the functions of interlining for leather garments. | 06 |

SECTION-B

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| 5(a) Define design and correlate it with product development. | 10 |
| 5(b) Sketch a lady's shoulder bag. Show the pattern transformation and adaptation techniques for the designed bag following pattern engineering. | 15 |

5(c)	Classify grading. Show the grading for a back panel of a garment basic block.	05
5(d)	How does enlargement differ from grading?	05
6(a)	Define basic color scheme. How do you specify a color?	06
6(b)	Show the application of a double split complementary color scheme in outfit designing.	11
6(c)	Clarify the terms: (i) Tone (ii) Shade and (iii) Tint. Illustrate the Munsell 5yR/5/10 numeric notation.	10
6(d)	Explain color saturation process. How do you calculate color saturation?	08
7(a)	Cite the elements of design. "Elements are the building blocks of a design"- Explain with examples.	10
7(b)	Illustrate the cycle time of fad, craze, and classic human dress patterns through proper diagram.	10
7(c)	Classify sewing machine. Mention the application of cylinder bed machine for leather goods assembling.	10
7(d)	Write down the applications of texture and line in leather products design.	05
8(a)	How do you adjust the pressure of the pressure foot of sewing machine?	08
8(b)	Show the flow diagram of new product development process. Sketch a needle and identify its different parts.	15
8(c)	What is meant by run in operation? Describe its necessity for the leather products industry.	07
8(d)	Distinguish among stitch, seam and sewing.	05

Department of Leather Engineering
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 B. Sc. Engineering 3rd Year 1st Term Examination-2022
LE 3101
Leather Manufacturing Technology IV

Time: 3.0 Hours

Full Marks: 210

- N.B. i) Answer any **THREE** questions from each section in separate scripts.
 ii) Figures in the right margin indicate full marks.
 iii) Assume reasonable data if any missing.

SECTION-A

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|------|--|----|
| 1(a) | Define leather drying. Describe the principle of energy and mass transfer during leather drying. | 10 |
| 1(b) | State the course of drying with the help of psychrometric chart. | 10 |
| 1(c) | Write on the constant drying rate period in leather drying. | 05 |
| 1(d) | Explain the mechanism of vacuum drying and toggle drying. | 10 |
| 2(a) | State the physical characteristics required for glove leather. | 08 |
| 2(b) | "Raw material selection can affect the quality of final leather"-Justify the statement. | 10 |
| 2(c) | Explain why conventional liming is not suitable for glove leather? | 06 |
| 2(d) | Write in detail to conduct liming operation during glove leather manufacturing. | 06 |
| 2(e) | State the properties of ideal lining leather. | 05 |
| 3(a) | Explain the liming process of glaze kid leather. | 09 |
| 3(b) | Why double bath process is preferred for chrome tanning operation? | 09 |
| 3(c) | Design a recipe for full shade dyeing of suede leather. | 10 |
| 3(d) | Describe the basic principle of book binding leather production. | 07 |
| 4(a) | Design a recipe for post tanning operation of glove leather. | 15 |
| 4(b) | Discuss the properties required for sulphur-oil-vegetable tanned picking band leather. | 10 |
| 4(c) | Write down the principle of bating for glaze kid leather production including recipe. | 10 |

SECTION-B

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|------|---|----|
| 5(a) | Explain the main purposes of leather finishing. | 08 |
| 5(b) | State different classifications of leather finishing. | 12 |
| 5(c) | Write short notes on: (i) Snuffing, (ii) Buffing, (iii) Stucco coat, (iv) Curtain coating finish and (v) Milling. | 15 |
| 6(a) | Write down the key properties of the following binders: (i) Albumin, (ii) Gelatin and (iii) NC lacquer. | 12 |
| 6(b) | Mention the highlighting points of selection of binders. | 08 |
| 6(c) | Write in detail why auxiliaries are used in leather finishing? | 10 |

6(d)	Mention some problems regarding deficient finish.	05
7(a)	Explain the properties of leather prior to finishing.	11
7(b)	Write down the differences between dye and pigments.	08
7(c)	Explain the synthesis of organic and inorganic dyes.	12
7(d)	State the necessity of pigments in leather finishing.	04
8(a)	"Pigments show high covering power"-Discuss the statement.	06
8(b)	Discuss the transformation of dyes to pigments.	12
8(c)	Describe the mechanism of cross-linkers and mention two cross-linkers with chemical structure.	09
8(d)	"Base coat is called the mother coat"-Justify the statement.	08

Department of Leather Engineering
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 B. Sc. Engineering 3rd Year 1st Term Examination-2022
LE 3107
Dyeing and Finishing I

Time: 3.0 Hours

Full Marks: 210

- N.B. i) Answer any **THREE** questions from each section in separate scripts.
 ii) Figures in the right margin indicate full marks.
 iii) Assume reasonable data if any missing.

SECTION-A

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|--|----|
| 1(a) Define metal complex dye (MCD) with mechanism of MCD with leather fibers. | 11 |
| 1(b) Explain the significance of color index (CI) with suitable example. | 08 |
| 1(c) Define azo dye. Why some of azo dyes are banned? | 05 |
| 1(d) State the formation of 1:2 metal complex dyes. | 11 |
| 2(a) "1:1 metal complex dye possesses better penetration than 1:2"- explain the statement. | 08 |
| 2(b) Write down the industrial routes of J-acid preparation. | 13 |
| 2(c) Write down the following reactions: (i) Beachmap reduction and (ii) Daw process. | 14 |
| 3(a) How will you manufacture dye from H-acid? | 12 |
| 3(b) Explain the oxidation and carboxylation reaction process. | 14 |
| 3(c) Prepare the following intermediates from Naphthalene: (i) Naphthalene-2- Sulphonic acid (ii) α - Naphthylamine (iii) Phthalic anhydride. | 09 |
| 4(a) Explain the effect of neutralization on dyeing. | 10 |
| 4(b) State the mechanism of quantisation of energy. | 08 |
| 4(c) "P-aminobenzene is yellow, but in acidic solution it becomes violet" -- explain the statement. | 08 |
| 4(d) State quinoid theory with its limitations. | 09 |

SECTION-B

- 5 Pigments are supposed to be insoluble and unaffected by any physical or chemical media providing some desired properties with particle size, shape, nature of surfaces, refractive indices or crystal nature. Covering properties of pigment make it useful while leather finishing stage.
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|--|----|
| (i) Explain how pigment provides an impact upon a substrate with its particle size, shape, nature of surface, refractive index or crystallinity? | 14 |
| (ii) "Covering properties of pigment make it more useful while leather finishing stage"- Prove the statement. | 08 |
| (iii) How pigments can be transformed as a neutral component?-Explain. | 13 |

- 6(a) Suppose you have supplied "D" grade loose grain leather. Now make a short design to prepare this leather for finishing operation. 10
- 6(b) Explain the function of refractive index in film formation. 10
- 6(c) Describe the formation process of finish film. 10
- 6(d) State the factors that affect the adhesion of finish film. 05
- 7(a) "Contact angle is important in finish film adhesion". Justify the statement. 10
- 7(b) Write down the mechanism of plasticization of finish film. 10
- 7(c) During or after finishing leather surface may get some troubleshooting. State the cause and prevention process of following problems- (i) Finish stick together (ii) Poor adhesion of finish to leather and (iii) Uneven spraying. 10
- 7(d) State the function of impregnation coat. 05
- 8(a) Make a recipe to produce aniline finish leather. 10
- 8(b) Briefly discuss the physical properties of a finish film. 10
- 8(c) What are the key factors for gloss retention? 05
- 8(d) "Melting points of the finishing materials play a vital role in conducting mechanical operations" – Explain. 10

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 B. Sc. Engineering 3rd Year 1st Term Examination-2022
LE 3115

Analytical Chemistry for Leather Manufacture

Time: 3 Hours.

Full Marks: 210

- N.B. i) Answer any **THREE** questions from each section in separate scripts.
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 iii) Assume reasonable data if any missing.

SECTION-A

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|------|---|----|
| 1(a) | State the following terms with necessary examples: (i) Accuracy and precision (ii) Qualitative analysis and quantitative analysis and (iii) Significant figure. | 09 |
| 1(b) | State the significance of Q-test. If the sulfide concentration of were found in the same pickle liquor 318, 319, 320, 322, 324 and 331 mg/L. One of the measured values was suspected. Apply Q-test and give your comment whether the suspected value is rejected or not at the 95% confidence level ($Q_{95}=0.625$) | 08 |
| 1(c) | Differentiate between determinate and indeterminate errors. | 04 |
| 1(d) | Describe a method in detail for chromium determination of your spent chrome liquor except instrumental methods. | 09 |
| 1(e) | With the help of a diagram describe the phase transition. | 05 |
| 2(a) | Define gravimetric analysis and volumetric analysis with examples. | 06 |
| 2(b) | Define the terms with examples-precipitate, precipitating agent, precipitant and precipitation. | 10 |
| 2(c) | Write down the properties of precipitate and precipitating agents. | 05 |
| 2(d) | Briefly discuss the instrumentation of thermogravimetric analysis with neat sketches. | 14 |
| 3(a) | Construe the thermal analysis, thermogravimetric analysis and thermogram. Write the application of thermal techniques in leather science. | 09 |
| 3(b) | List out the factors that affect in thermal analysis. Discuss any three of them. | 11 |
| 3(c) | Neat sketches the ion chromatography (IC). Briefly discuss the mobile phase, eluent pump and stationary phase. | 15 |
| 4(a) | Define the terms-chromatography and ion chromatography. Briefly discuss the principle of ion separation with suitable figure. | 08 |
| 4(b) | Describe the selection criteria of column in terms of ion chromatography. | 12 |
| 4(c) | Define retention time. If the calibration curve for the sulfate (SO_4^{2-}) solution by ion chromatography was found $y=0.1209x-0.0031$ and tannery wastewater shows the response at the same retention time 1.5927 unit. Calculate the sulfate concentration of the tannery wastewater. | 05 |
| 4(d) | State working principle of gas chromatography with schematic illustration. | 10 |

SECTION-B

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|------|--|----|
| 5(a) | Distinguish between atomic and molecular spectroscopy. Why is atomic absorption spectroscopy (AAS) limited to metal detection? | 08 |
| 5(b) | State the basic principles of atomic absorption spectroscopy (AAS) with neat sketches of the instruments. | 12 |
| 5(c) | Write down the construction, working procedure, advantages and limitation of pre-mix burner (PCB) used in atomizer of AAS with proper sketches. | 10 |
| 5(d) | Mention some strength and weakness of AAS. | 05 |
| 6(a) | Schematically show the instrumentation of the modern HPLC system. | 08 |
| 6(b) | Why degassing of mobile phase is necessary in HPLC? Write on the vacuum degasser with neat sketches. | 10 |
| 6(c) | Elucidate the different stages of manual sample injection system in HPLC with schematic representation. | 09 |
| 6(d) | Differentiate between RP-HPLC and NP-HPLC systems. | 08 |
| 7(a) | Explain the modes of Electronic Transition (ETT) between different molecular orbital occur in UV-Vis spectroscopy. | 10 |
| 7(b) | Write down the construction and working procedure of monochromator used in UV-Vis spectroscopy with schematic diagram. | 10 |
| 7(c) | A solution of any dye "D" has an absorbance at 480nm of 0.64 in a 0.5cm length cuvette. The straight line equation of the calibration for the dye is given as $y = 0.0078x + 0.0012$. Calculate the concentration of the dye. [Concentration of dye for standard solution was measured as mg/L] | 05 |
| 7(d) | Illustrate the instrumentation of mass spectroscopy with neat sketches. | 10 |
| 8(a) | Write down the sample cell and sample preparation techniques with IR spectroscopy. | 10 |
| 8(b) | Briefly discuss the Michelson Interferometer in FTIR spectroscopy with schematic representation. | 10 |
| 8(c) | Find the wave number of IR radiation required for the N=O bond [Atomic mass of nitrogen and oxygen are 2.326×10^{-23} g and 2.658×10^{-23} g, respectively. Speed of light $c = 2.99 \times 10^{10}$ cms ⁻¹] | 05 |
| 8(d) | State the following aspects of NMR spectroscopy with proper examples: (i) Peak number (ii) Peak shifting and (iii) Peak splitting | 10 |